



Effect of TiN on the microstructure and mechanical property of As-cast TiZrNbVAl lightweight high entropy alloy

Jiaxiang Cui^a, Qian Zhao^a, Bang Dou^a, Yuxue Liu^a, Yaqiong Fu^a, Shi-Hai Sun^{a,b,*},
Liang Wang^a, Hongnian Cai^a, Yunfei Xue^{a,b}

^a School of Materials Science and Engineering, Beijing Institute of Technology, Beijing 100081, China

^b Tangshan Research Institute, BIT, Tangshan 063000, China

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ABSTRACT

The grain size of as-cast light weight high entropy alloy (LHEA) is required to be refined and its mechanical property should be improved further to satisfy its engineering application. In this paper, we investigated the effect of TiN on the microstructure and mechanical property of as-cast TiZrNbVAl LHEA with body-centered cubic (BCC) structure. TiN can serve as the potent nucleating agent, and reduce the grain size of as-cast ingot from 275 μm to 97 μm with 5 wt% addition. Part of TiN would dissolve during melting and reprecipitate again in the plate-like shape ($\sim 0.3 \mu\text{m}$ thickness) at the grain interiors and grain boundaries during solidification, which could increase the strength but decrease the ductility. 3 wt% TiN addition can increase the yield strength of TiZrNbVAl LHEA near two times, from 804 MPa to 1603 MPa and maintain $\sim 25\%$ compression ductility. The significant strength improvement is ascribed to the synergy strengthening effect of TiN, interstitial N atom and refined grains. Our work is a valuable reference for the improvement of mechanical properties of as-cast BCC LHEAs.

1. Introduction

Lightweight high-entropy alloys (LHEAs) with body-centered cubic (BCC) structures have received extensive attention due to their high specific strength [1–3], high elevated temperature strength [4], good corrosion [5] and wear resistance [6]. TiZrNbVAl LHEA system usually exhibits considerable strength and ductility owing to the solid solution strengthening effect and low valence electron concentration [7,8], which are promising candidates for the applications in the field of aerospace, automobile, and marine.

Nevertheless, the exploration of BCC-LHEAs with superior mechanical properties is still a hot research topic. To date, several strengthening strategies have been applied to enhance the strength of BCC-HEAs. For example, the solid solution strengthening resulting from atomic radius mismatch [9,10], the precipitation strengthening caused by the introduction of the second phase [11,12], and the grain boundary strengthening following deformation and recrystallization treatments [13,14]. However, these strengthening methods have a limited strength improvement on BCC-LHEAs, generally less than 300 MPa, which limits their application in fields demanding high strength.

Recent studies have shown that the addition of ceramic particles, such as carbide [15], nitride [16] and oxide [17] can significantly improve the strength of the HEAs. Wang et al. [18] increased the yield strength from 1250 MPa to 2500 MPa by adding Al_2O_3 particles to HfNbTaTiZrV refractory high entropy alloy (RHEA), which was nearly doubled. The increase of yield strength is generated by multiple strengthening mechanisms, including solid solution strengthening, interstitial strengthening, grain boundary strengthening and dispersion strengthening. However, the incoherent interface between Al_2O_3 and the matrix is easy to cause stress concentration around the precipitate, leading to a sharp decrease in the ductility (from 33 % to 6 %). To solve this problem, Wang et al. [19] added HfN to HfNbTaTiV HEA, which has a semi-coherent interface with the HfNbTaTiV RHEA matrix. The strength of the alloy could increase by nearly two times, and the ductility is still maintained well ($>10\%$). Therefore, adding the ceramic phase having low lattice mismatch with the matrix is an effective approach to improve the strength of BCC-LHEAs and not to scarify its ductility significantly.

In addition, the ceramic phase having low lattice mismatch with the matrix could act as an effective nucleating agent to refine the grain size

* Corresponding author at: School of Materials Science and Engineering, Beijing Institute of Technology, Beijing 100081, China.

E-mail address: shihaisun@bit.edu.cn (S.-H. Sun).

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of the cast alloy and improve its mechanical property [20]. This is because the interface energy of the low lattice mismatch interface is low, which is beneficial to the heterogeneous growth, increasing the nucleation rate [21]. Casting is a low-cost method to produce the large-scale component, but its usage is severely restrained by the large grain size [22]. Therefore, adding a proper low lattice mismatch ceramic phase is of great significance for the preparation of high-performance large-scaled LHEA components through casting. Until now, the grain refinement of as-cast HEAs is mainly to add appropriate elements to increase the excessive composition [18,23,24]. The research about the nucleation agent in HEA is quite rare, and its influence on the grain refinement and mechanical property should be investigated in depth.

In this work, non-equimolar TiZrNbAl BCC-LHEA system was selected as the model alloy because of its high specific yield strength, excellent ductility, and excellent phase stability, which had been indicated in our previous reports [3]. The as-cast structure of this LHEA has a wide application prospect in many engineering components, such as aircraft rudder transmission brackets, shielding plates and wings. Then, TiN was selected as the low lattice mismatch ceramic phase in it to refine the grain size and strengthen the alloy (Supplementary Note 1 and Fig. S1 of Supplementary materials.). The effect of TiN on the microstructure and mechanical property of TiZrNbAl LHEA were analyzed in detail and the grain refinement and strengthening mechanism were discussed. The results in this study could be a good reference for the preparation of as-cast BCC LHEAs with high mechanical property.

2. Experimental procedures

Non-equiatomic TiZrNbAl LHEA with different TiN addition (0, 3, and 5 wt%) were prepared by the levitation melting furnace in argon atmosphere. Hereafter, they were named as TiN-0, TiN-3 and TiN-5 alloy, respectively. Because TiN would decompose to generate N atom in TiZrNb-based HEAs [19,25,26], and the solid solubility limit of N in Zr is the smallest among the binary phase diagram of N and metal elements of the investigated composition in this study, which is 4.4 at% (Fig. S2 of the Supplementary materials), we designed alloys with TiN concentrations greater than 4.4 at% (5 wt% in the experiment) to keep more TiN particles in the alloy. For comparison, we also designed the content of TiN less than 4.4 at% (3 wt% in the experiment). The raw material purity of Ti, Zr, Nb, V, and Al was more than 99.95 wt%, and that of TiN was and the size of TiN particle (The purity was more than 99 %) was ~ 300 nm. In order to ensure the elemental uniformity of the ingot, each ingot was melted three times, and the alloy ingot was flipped before each melting. Then, TiN was uniformly put to the surrounding of the ingot and the ingot was melt again and hold for 30 s followed by furnace cooling. The ingot had a diameter of 50 mm and a height of 50 mm. The oxygen concentration in the alloy was tested by G8 Galileo ON/H analyzer, and the actual oxygen concentration of TiN-0, TiN-3 and TiN-5 LHEAs is shown in Table S1 of Supplementary materials. The results show that only a small amount of oxygen impurities is introduced in the melting process, which has a slight effect on the microstructure and mechanical properties of the alloy.

The distribution and morphology of TiN in the TiZrNbAl LHEA were analyzed by scanning electron microscopy (SEM, HITACHIS4800), and the grain morphology and crystal orientation were analyzed by electron backscatter diffraction (EBSD). The element distribution was analyzed by energy dispersive spectroscopy (EDS) detector. The constituent phases were detected by X-ray diffraction (XRD, Bruker AXS D8 Advance diffractometer) using Cu-K ray in a diffraction angle (2θ) range of $20\text{--}100^\circ$, and 2θ -scanning speed of $0.2^\circ/\text{s}$. The hardness of alloys was measured by a LECO LM700AT Vickers hardness instrument. The $\Phi 4\text{ mm} \times 8\text{ mm}$ cylindrical specimens were used for compressive test. The compressive tests were carried out on MTS electro-hydraulic servo universal testing machine at room temperature with strain rate of $1 \times 10^{-3}\text{ s}^{-1}$. The samples are tested at least three times to ensure data reliability and stability.

3. Results and discussion

3.1. Microstructural characterization

Fig. 1 shows the XRD pattern of as-cast TiZrNbAl LHEA with different TiN contents. As shown in Fig. 1(a), all the alloys exhibited the strong diffraction peak of BCC structure, which stemmed from the matrix. In TiN-3 and TiN-5 alloys, only the weak diffraction peaks (Fig. 1(b)) from TiN with FCC structure could be detected beside those from BCC matrix. In order to accurately analyze the crystal structure of TiN-0, TiN-3 and TiN-5 alloys, Fig. S3 of Supplementary materials shows the TEM bright-field images and selected area electron diffraction of the three alloys, and the results show that the lattice constants of TiN-0, TiN-3 and TiN-5 were found to be $a = 3.301\text{ \AA}$, $a = 3.324\text{ \AA}$ and $a = 3.335\text{ \AA}$, respectively, indicating that the lattice constants increase after the addition of TiN. This was caused by the fact that after TiN decomposition, a part of Nitrogen occupies the gap position of the alloy and thus increases the lattice parameters of the LHEAs. It should be noted that TiN could decompose during the melting process, because its decomposition temperature (1050°C [25]) is lower than the melting temperature ($\sim 1650^\circ\text{C}$) of the alloy. This means that the decomposed N may combine with other reactive elements in alloy, such as Zr and Ti. However, no new phase was detected by XRD, which suggests that the decomposition of TiN does not generate new phases. Table S2 of Supplementary materials shows the mixing enthalpy of the atomic pairs in TiZrNbAl-TiN LHEAs, and the results show that the mixing enthalpies between N and four metallic elements are all negative, and the mixing enthalpy between N and Ti/Zr is the most negative. Moreover, Fig. S2 (a-b) of Supplementary materials shown that at high temperature ($>2000^\circ\text{C}$), compared with ZrN, the concentration of N required for TiN is smaller, and the concentration of Ti (50 at%) in TiZrNbAl LHEA is much higher than that of Zr (18 at%), which leads to the decomposed N may dissolve in the matrix or combine with Ti again during solidification process.

Fig. 2 shows the microstructure in TiZrNbAl LHEAs with different TiN contents. As shown in Fig. 2(a-c), all the specimens exhibited the equiaxed grains, and the grain size decreased with increasing the TiN content. The precipitates with two different sizes could be found in TiN-3 and TiN-5 alloys. The larger precipitates (indicated by the light blue arrows in the insert images of Fig. 2(b-e)) were near spherical shape, whose diameter was $\sim 5\text{ }\mu\text{m}$, and mainly existed at the grain interiors. The elemental line scan (yellow line in Fig. 2(d)) across them (Fig. 2(f)) shows they were rich in Ti and N, which suggests they were artificially added TiN particles. It should be noted that their sizes were much larger than those of originally added TiN particles (Fig. S4 of Supplementary materials), suggesting the reprecipitated TiN may preferentially grow on the original TiN particles and make them grow bigger. The EDS point analysis shows that the N content of the TiN-3 and TiN-5 alloys was 2.0 at% and 3.4 at%, respectively. This suggests that part of N still exist in the matrix, which is consistent with the results of Fig. S3 of Supplementary materials. The fine precipitates (indicated by the red arrows in Fig. 2(d, e)) were plate-like shape, whose thickness was $\sim 0.3\text{ }\mu\text{m}$, and they could be found at the grain interiors and grain boundaries (Fig. S5 of Supplementary materials shows the statistical results of the size of the fine precipitates). These precipitates were also rich in Ti and N, thus they should be the reprecipitated TiN during solidification.

The TEM analysis for the nano sized precipitates is shown in Fig. 2(g-h). As shown in Fig. 2(g), the precipitates with the length of $100\text{--}300\text{ nm}$ were observed in the TEM bright-field image (Fig. S6 of Supplementary materials highlights the shape and size of the precipitates). The selected area electron diffraction (SAED) analysis of the precipitates was carried out. As shown in Fig. 2(g), the results indicate in addition to the SAED with BCC structure, the SAED with FCC structure was also observed. The lattice constants of BCC and FCC structures are $3.33\text{ \AA}$ and $4.23\text{ \AA}$, which are similar to those of LHEA and TiN, respectively. Therefore, combined with the composition analysis of point A in Fig. 2(g), it could be

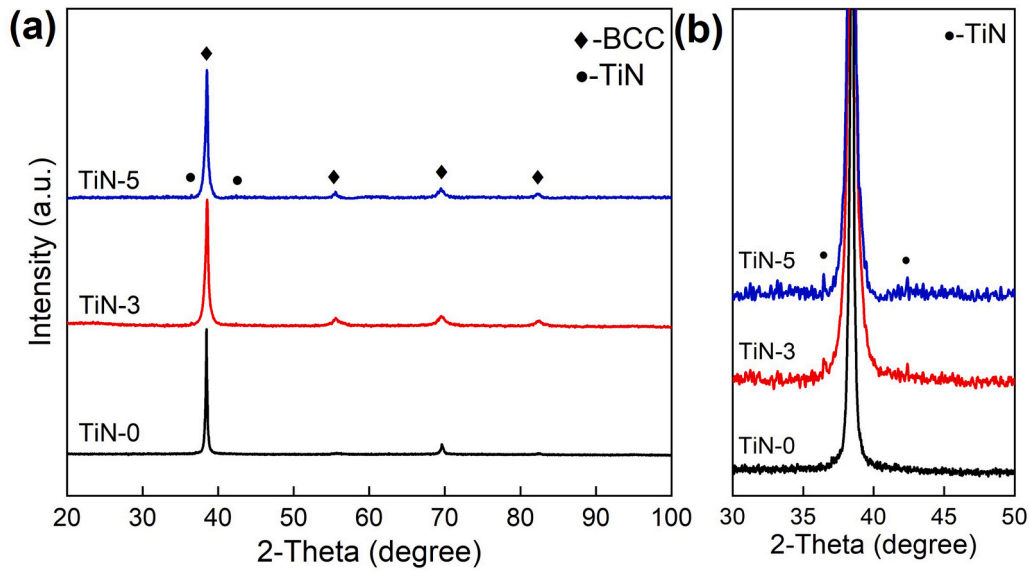


Fig. 1. (a) XRD patterns of TiZrNbVAl LHEA with different TiN content; (b) Enlarged XRD patterns ranging from 30° to 50°.

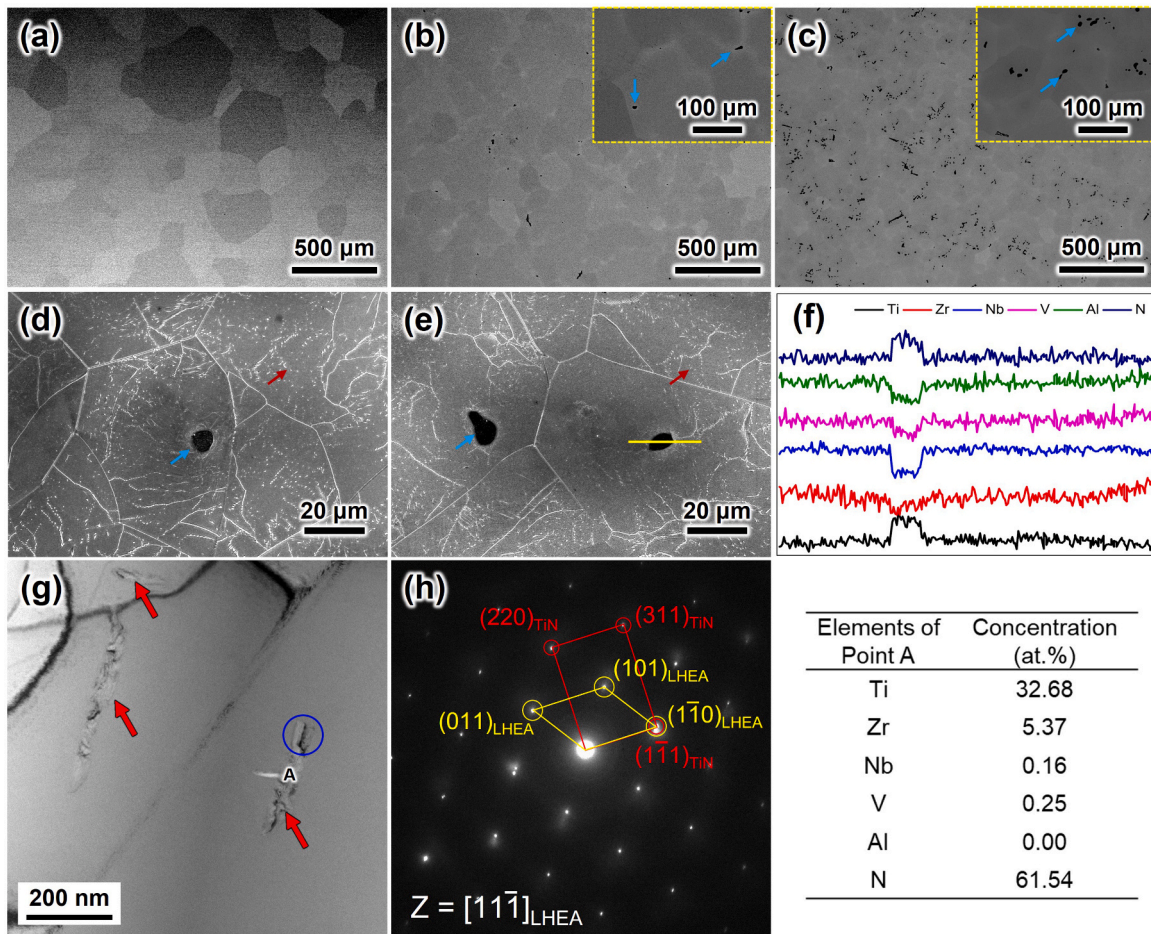


Fig. 2. Microstructure of TiZrNbVAl LHEA with different TiN contents: (a) TiN-0; (b, d) TiN-3; (c, e) TiN-5. (f) EDS line scan result of the black precipitate crossed by the yellow line in e panel. (g) TEM bright-field image of TiN-5 LHEA showing the nano sized precipitates; (h) The SAED pattern of blue circle in g panel.

confirmed that the nano sized precipitates are TiN. Moreover, it could be seen in Fig. 2(g) that TiN has the following orientation relationship with TiZrNbVAl LHEA:

$$\langle \bar{1}12 \rangle_{\text{TiN}} \parallel \langle 11\bar{1} \rangle_{\text{LHEA}}, \quad (1\bar{1}1)_{\text{TiN}} \parallel (1\bar{1}0)_{\text{LHEA}}$$

This suggests that TiN is coherent with the BCC matrix.

Fig. 3 shows the inverse pole figure map of TiZrNbVAl LHEA with different TiN contents. In order to exclude the influence of sampling position on grain size, we select the same position of ingot to carry out

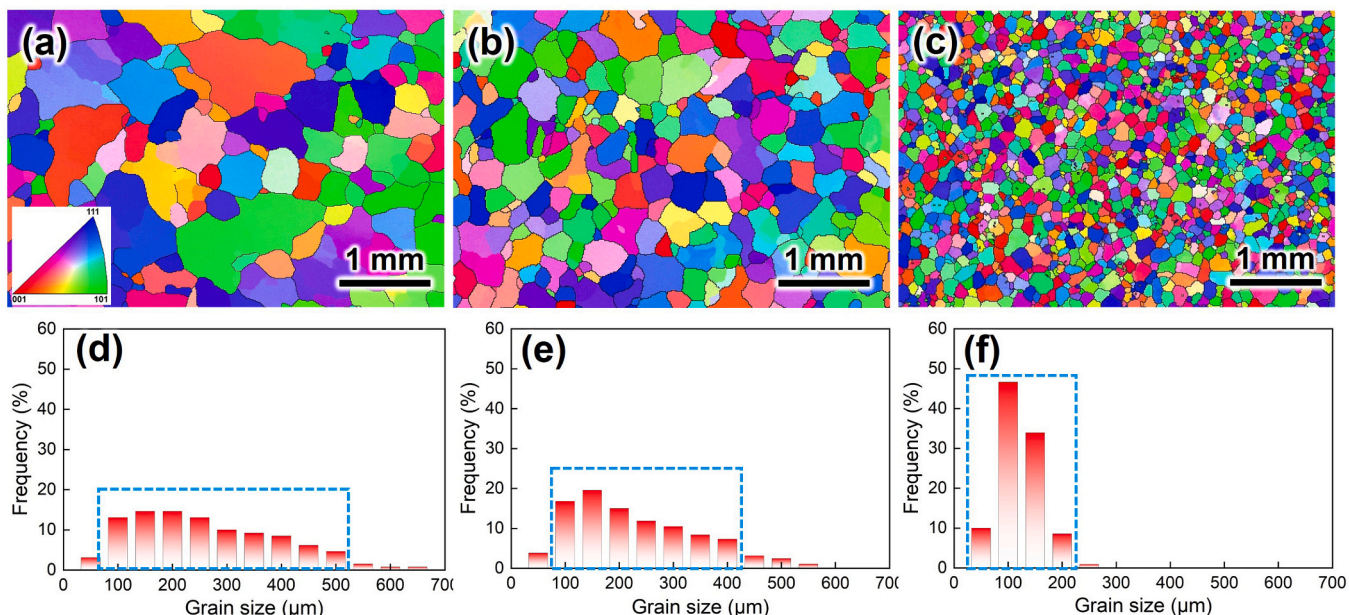


Fig. 3. The EBSD inverse pole figure maps and grain size distributions of TiZrNbVAl LHEA with different TiN contents: (a, d) TiN-0, (b, e) TiN-3, (c, f) TiN-5.

grain size statistics for LHEAs with different TiN content. As shown in Fig. 3(a-c), it was equiaxed crystals with random orientation in all the alloys. The grain size decreased 35 %, from $\sim 275 \mu\text{m}$ to $\sim 97 \mu\text{m}$, with increasing TiN content from 0 to 5 wt%. Moreover, as shown in Fig. 3(d-f), the grain size in LHEAs with different TiN content was statistically analyzed. As shown in Fig. 3(d), the size of the most grain in TiN-0 LHEA is between 100 and 500 μm (the sum content is about 94 %), and the grain size greater than 500 μm is greater than 3 %. However, the size of the most grain in TiN-3 LHEA is between 100 and 400 μm (the sum content is about 90 %), and the grain size greater than 500 μm is less than 1 %, indicating that there is almost no large size grains ($>500 \mu\text{m}$) after adding TiN. Finally, when the TiN content is 5 wt%, the size of the most grains in the LHEAs is between 100 and 200 μm (the sum content is about 99 %), and the grain size greater than 300 μm is 0, that is, TiN-5 LHEA is uniform and fine equiaxed crystal. This suggests that TiN become the effective heterogenous nucleation site, and significantly refine the grain size of TiZrNbVAl LHEA.

3.2. Mechanical properties

Fig. 4 shows the compressive stress strain curves of TiZrNbVAl LHEA

with different TiN contents at room temperature and its mechanical property comparison with other reported LHEAs. Firstly, as shown in Fig. 4(a), with the increase of TiN content, the yield strength of the LHEA gradually increases, but the plasticity of the LHEA decreases. The indentation results of hardness test for LHEA with different TiN contents (Fig. S7 of the Supplementary materials) also indicate that plastic deformation becomes difficult with the increase of hardness after adding TiN.” was added. The yield strength of TiN-0 is 803 MPa and the compression strength is 2274 MPa. When the TiN content is increased to 3 wt%, the yield strength is increased to 1603 MPa, an increase of 2 times. However, compared with TiN-0, the compression ductility decreased from more than 50 % to $24.6 \pm 3.2 \%$, which was reduced by more than half. Furthermore, the yield strength of TiN-5 is 2124 MPa, which is almost 3 times larger than that of the original alloy. However, due to the segregation of TiN at grain boundaries, the compression ductility of TiN-5 LHEA is limited, only $2.7 \pm 0.4 \%$. Secondly, as shown in Fig. 4(b), among the alloys with different TiN contents, TiN-3 alloy had the best strength and ductility matching, which superior to most of the other reported LHEAs. This suggests that adding low lattice mismatching particle is a promising strategy to improve the mechanical property of the LHEAs.

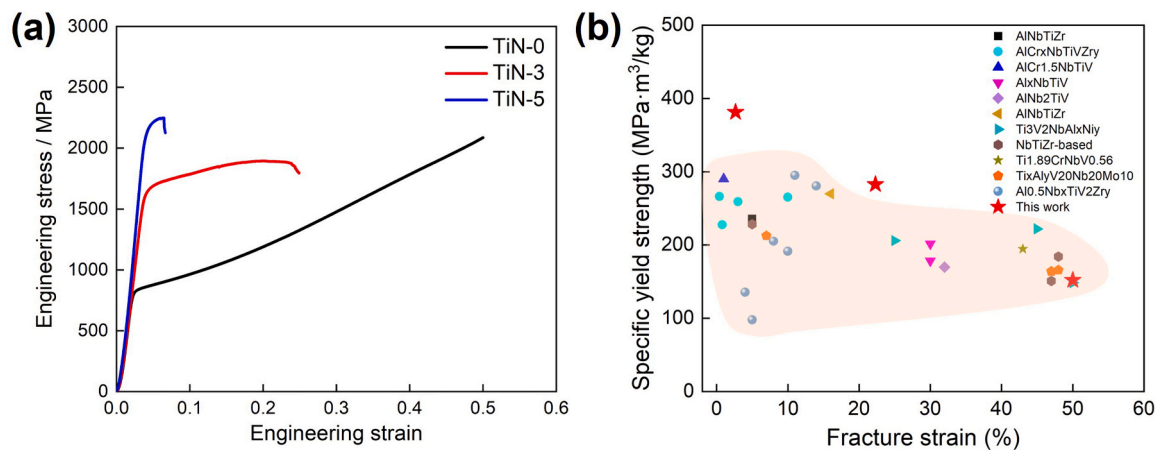


Fig. 4. (a) Compressive stress-strain curves of TiZrNbVAl LHEA with different TiN contents under strain rate of $1 \times 10^{-3} \text{ s}^{-1}$ at room temperature. (b) Compressive property comparison between TiZrNbVAl LHEAs with different TiN contents and other LHEAs [27–37].

Fig. 5 shows the fracture propagation of TiZrNbVAl LHEA with different TiN contents after compression deformation. For TiN-0 alloy, the sample only become the drum shape after compression, and no crack was observed (Fig. 5(a)), the deformation band was mainly along the direction with 45° deviation from the compression direction. However, TiN-3 and TiN-5 LHEAs were broken after deformation, and the crack was mainly along the direction with 45° deviation from the compression direction (Fig. 5(b, c)). In TiN-3 alloy, it can be observed that the crack initiated from the TiN particle (blue arrow in Fig. 5(b)). In TiN-5 alloy, the fracture surface exhibited cleavage like shape (Fig. 5(c)), indicating that intergranular fracture occurs during compression. This should be because the grain boundary binding force is decreased by the TiN at grain boundaries.

Fig. 6 shows the TEM bright field images of TiN-3 alloy after deformation. In Fig. 6(a), there were abundant dislocation entanglement and plugging around the grain boundaries, which suggests that the grain boundaries impede the dislocation motion. In regions far from the grain boundaries, the dislocations slip in the plane and form the dislocation networks. The grain size in TiN-3 alloy is decreased by TiN addition, thus its grain boundary density is higher than that in TiN-0 alloy. This could result in more dislocation entanglement and plugging at the grain boundaries of TiN-3 alloy, leading to its yield strength increase. In addition, it is worth noting that interactions between dislocations at the interface play a predominant role in the mechanical properties. As shown in Fig. 6(b-c), when the dislocation arrays meet the precipitates of TiN, the propagation of the dislocation in the arrays is hindered, which increases the resistance of the dislocation movement. Moreover, these randomly distributed dislocations become entangled around the precipitated phase further enhancing resistance against their movement and increasing the yield strength. This is another reason for the yield strength increase in TiN-3 alloy.

3.3. Effect of TiN on grain size

Here, the edge-to-edge matching (E2EM) model is used to analyze the effect of TiN on the grain refinement of TiZrVNbAl LHEA, which has been often used to evaluate the grain refinement effect of the nucleating agent in Aluminum and Magnesium alloys [38,39].

The E2EM calculation involves the following steps: (1) Identification of matching row pairs and matching plane pairs, typically aligned along the close-packed (CP) directions and planes, respectively. (2) Calculation of atomic spacing mismatch (f_r) for matching row pairs and plane spacing mismatch (f_d) for matching plane pairs; (3) If atomic spacing mismatch satisfy $f_r < 10\%$ and plane spacing mismatch satisfy $f_d < 6\%$, it is considered that foreign additives can become the potent nucleating agent for the matrix alloy.

The detailed crystallographic data used for calculating of f_r and f_d can be found in Table 1, and f_r and f_d of some typical nucleation agents and TiZrVNbAl LHEA are compared in Fig. S8 of Supplementary materials. It can be seen that only the atomic and plane mismatches between TiN and TiZrVNbAl LHEA are less than the upper limit for the high potency nucleating agent in the E2EM model at the same time. This suggests that TiN is good nucleation agent for TiZrVNbAl LHEA, and its grain

refinement will be superior to the other commonly used nucleation agent.

To better understand the grain refinement mechanisms in this study, Fig. 7 shows the schematic illustration for the nucleation and growth processes of as-cast TiZrVNbAl LHEA with and without TiN addition. In this diagram, the circle represents the nucleating grain, and T_E and T_A are the equilibrium liquidus temperature and actual temperature profile ahead of the advancing front of the nucleating grain, respectively. The region between T_E and T_A is characterized by constitutional supercooling. ΔT_n and ΔT_{cs} are the critical undercooling and the constitutional supercooling, respectively, and their amount are qualitatively represented by the length of red and orange bars. The nucleation grain will continue to grow until ΔT_{cs} is equal to ΔT_n [40], where the next nucleation event occurs, and the constitutional supercooling region needs to be re-established to enable the next nucleation. Therefore, the average grain size (d_{gs}) of the alloy is governed by ΔT_n . In TiZrVNbAl LHEA without TiN addition (Fig. 7(a)), the critical undercooling ΔT_{n1} required for the nucleation is larger, then the grain needs to grow larger to establish the large enough constitutional supercooling. However, when TiN is added to TiZrVNbAl LHEA (Fig. 7(b)), ΔT_{n2} required for nucleation will be lower because TiN has good crystallographic matching with TiZrVNbAl LHEA and could provide potent heterogeneous nucleation site [41]. Then the grain only needs to grow a little to form the required constitutional supercooling. More TiN particles could provide more heterogeneous nucleation site, which will increase the nucleation rate and make the grain size become finer. However, it is noteworthy that only a small proportion of nucleation agent is needed for grain refinement, and excessive nucleation agent will not generate better grain refinement effect [42] and may deteriorate the ductility of the alloy [43].

3.4. Effect of TiN on strengthening

It is surprising that the yield strength of the TiZrNbVAl LHEA increases three times with only 5 wt% TiN addition. Therefore, it is necessary to clarify the strengthening mechanism of TiZrNbVAl LHEA with TiN addition.

The contribution of different strengthening mechanism on the yield stress of the alloy can be expressed as [44]:

$$\sigma_{YS} = \sigma_m + \Delta\sigma_{ss} + \Delta\sigma_G + \Delta\sigma_N + \Delta\sigma_p \quad (1)$$

where, σ_m is the yield strength of the matrix, and $\Delta\sigma_{ss}$, $\Delta\sigma_G$, $\Delta\sigma_N$ and $\Delta\sigma_p$ are the yield strength increment caused by solid solution strengthening, fine-grained strengthening, interstitial-solution strengthening of N and second phase strengthening of TiN, respectively.

σ_m can be calculated by the following equation:

$$\sigma_m = \sum \sigma_{yi} X_i \quad (2)$$

where, σ_{yi} and X_i are the yield strength and atomic fraction of each component element i , respectively.

$\Delta\sigma_{ss}$ can be expressed as:

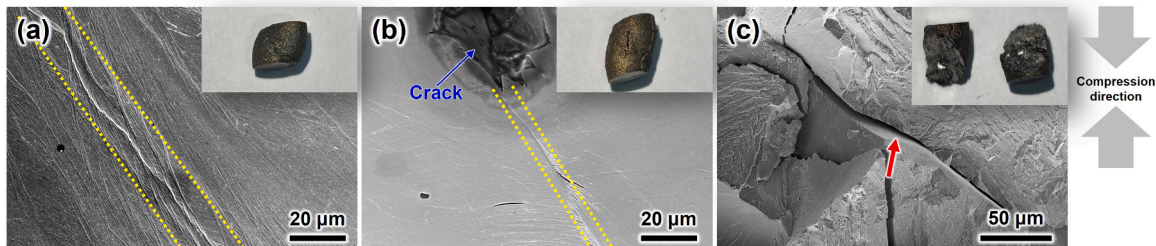


Fig. 5. Fracture morphologies of TiZrNbVAl LHEAs with different TiN contents.

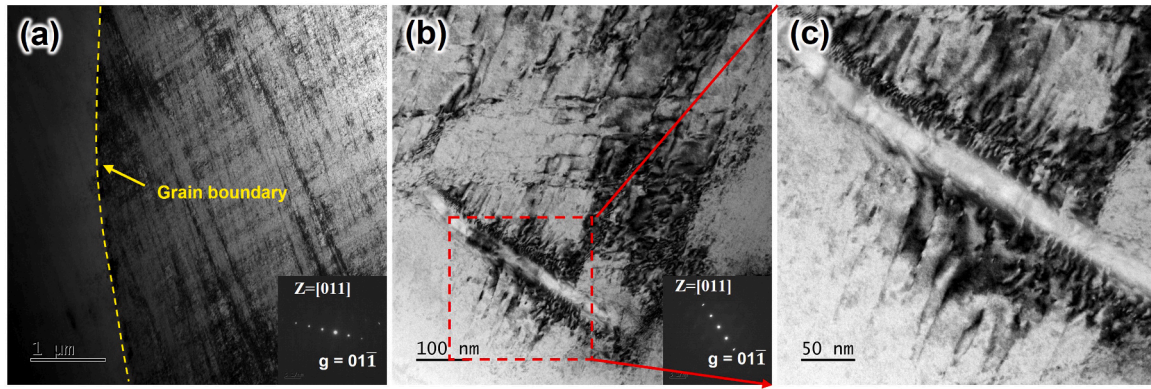


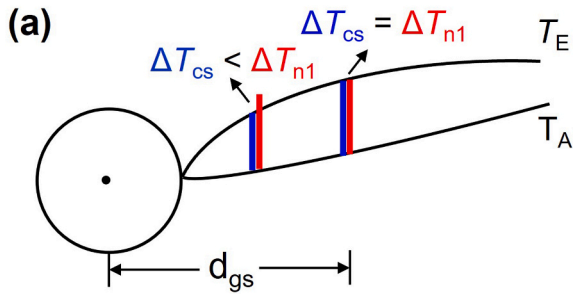
Fig. 6. TEM bright field images of TiN-3 alloy after compressive deformation: (a) near grain boundary, (b, c) near TiN precipitates.

Table 1

The detail crystallographic data used for the calculation of f_r and f_d for some typical nucleation agent and TiZrNbAl LHEA.

Phases	Structure	CP rows	d_{uvw} (nm)	f_r	CP planes	d_{hkl} (nm)	f_d
TiC	FCC	$\langle 110 \rangle$	0.306	6.99	$\{111\}$	0.249	6.41
TiB ₂	HCP	$\langle 2\bar{1}\bar{1}0 \rangle$	0.303	5.94	$\{1\bar{1}00\}$	0.324	38.46
TiN	FCC	$\langle 110 \rangle$	0.299	3.50	$\{111\}$	0.244	4.27
Y ₂ O ₃	BCC	$\langle 110 \rangle$	0.325	13.45	$\{220\}$	0.265	13.41
TiZrNbAl	BCC	$\langle 111 \rangle$	0.286	—	$\{110\}$	0.234	—

Without TiN addition



With TiN addition

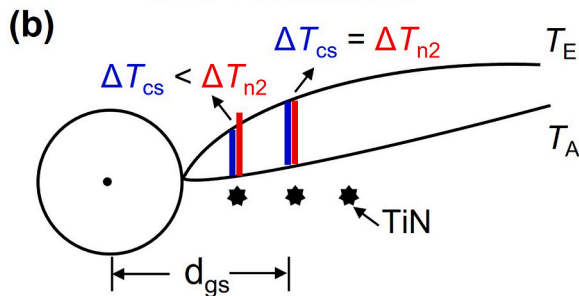


Fig. 7. Schematic illustration of the effect of TiN on the nucleation and growth of as-cast TiZrNbAl LHEA (a) with and (b) without TiN addition.

$$\Delta\sigma_{ss} = G \left(\sum \varepsilon_i^2 X_i \right)^{2/3} / 45 \quad (3)$$

where, G is the shear modulus of the alloy, which can be estimated by the following equation for LHEA:

$$G = \sum G_i X_i \quad (4)$$

where, G_i is the shear modulus of component element i .

ε_i is the mismatch parameter of the component element i , which can be calculated as follows:

$$\varepsilon_i = (\eta_i^2 + \alpha^2 \delta_i)^{1/2} \quad (5)$$

where, α describes the differences in the interaction forces between edge and screw dislocations with foreign atoms, and it could be taken as 16 for the BCC alloy [44]. η_i and δ_i are the average atomic modulus and size difference between the i -th element and other elements, respectively. They can be calculated with the following equations:

$$\eta_i = \frac{9}{8} \sum X_j \eta_{ij} \quad (6)$$

$$\delta_i = \frac{9}{8} \sum X_j \delta_{ij} \quad (7)$$

where $\eta_{ij} = 2(G_i - G_j)/(G_i + G_j)$ and $\delta_{ij} = 2(r_i - r_j)/(r_i + r_j)$ are the shear modulus and atomic radius difference between the i -th and j -th elements, respectively. The σ_{yi} , r_i , and G_i of each element are list in Table S3 of supplemental materials.

The effect of grain size on the yield strength can be expressed by the Hall-Petch equation:

$$\Delta\sigma_G = Kd^{-1/2} \quad (8)$$

where K is the strengthening constant, which is taken as 288.3 MPa·μm^{-1/2} for TiZrNbAl LHEA according to the previous study [45], and d is the average grain size.

The interstitial solid solution strengthening effect can be calculated by [25,46–48].

$$\Delta\sigma_N = kc^{2/3} \quad (9)$$

where, k is the strengthening factor related to the material, which is taken as 250 MPa in BCC-HEAs [25], and c is the concentration of the interstitial element.

The increment of yield strength caused by the second-phase particle ($\Delta\sigma_p$) can be calculated by:

$$\Delta\sigma_p = \Phi \frac{Gb}{d_p} \left(\frac{6f_p}{\pi} \right)^{\frac{1}{3}} \quad (10)$$

where, Φ is the aspect ratio of the second-phase particles, which is taken as the average value of the larger and smaller particles in this study, respectively. b is the Burgers vector [2], d_p is the average particle size, f_p is the content of the second-phase particles inside the grains. The parameters used for the calculation of second-phase strengthening of TiN are listed in Table 2.

Fig. 8 shows the percentage contribution of different strengthening mechanism to the yield stress of TiZrNbVAl LHEAs with different TiN contents. The calculated results agree well with the experimental results, indicating the credibility. It can be seen that it is the second phase strengthening of TiN and the interstitial solid solution strengthening of N who contribute a lot to the strength enhancement of TiZrVNBAl LHEA after TiN addition. TiN will decompose during melting and part of it will reprecipitate on the grain boundaries during solidification. Therefore, although increasing the addition content of TiN could increase the alloy strength, it will increase the amount of reprecipitated TiN on the grain boundaries at the same time and decrease the ductility of alloy greatly. In order to utilize the strengthening effect of TiN, it is necessary to decrease the decomposition of TiN and inhibit its precipitation on the grain boundaries by optimizing the addition way of TiN and the processing parameter in the future study. It is noteworthy that the interstitial N atoms increase the strength of TiZrVNBAl prominently, and it could remain the good ductility based on the interstitial solid solution strengthening mechanism. Thus, N addition represents a promising strategy for improving the mechanical properties of TiZrVNBAl LHEA.

4. Conclusions

In this study, the effects of TiN contents on the microstructure and mechanical properties of TiZrNbVAl LHEAs were systematically studied. The results show a gradual reduction in grain size of the as-cast TiZrNbVAl LHEA with increasing TiN contents. Specifically, when the TiN content is 5 wt%, the grain size decreases from 275 μm to 97 μm . According to the E2EM model, TiN acts as a heterogeneous nucleation agent for as-cast TiZrNbVAl LHEA, promoting nucleation rate and refining grain size during solidification. The incorporation of TiN simultaneously induces interstitial strengthening, grain boundary strengthening and dispersion strengthening mechanisms that enhance the strength of TiZrVNBAl LHEA from 803 MPa to 1603 MPa after adding 3 wt% TiN, while maintaining ~25 % compressive ductility. The analysis of the strengthening mechanism reveals that interstitial solid solution of N atoms increases the lattice distortion and exhibits excellent interstitial strengthening effect. Moreover, TiN can not only refine and improves grain boundary strengthening effect, but also hinders the movement of dislocations, which significantly improves the strength. Our findings provide valuable data and unique insights into refining grain refinement techniques and enhancing mechanical properties of as-cast BCC-LHEAs.

CRediT authorship contribution statement

Yuxue Liu: Methodology, Investigation, Formal analysis. **Bang Dou:** Methodology, Investigation, Formal analysis. **Qian Zhao:** Investigation, Formal analysis. **Hongnian Cai:** Methodology, Investigation, Formal analysis. **Liang Wang:** Investigation, Formal analysis. **ShiHai Sun:** Writing – review & editing, Methodology, Investigation, Formal analysis. **Yaqiong Fu:** Methodology, Formal analysis. **Jiaxiang Cui:** Writing – original draft, Methodology, Investigation, Formal analysis, Data curation. **Yunfei Xue:** Writing – review & editing, Investigation, Funding acquisition.

Table 2

Parameters used for the calculation of the second phase strengthening of TiN.

parameter	Φ	d_p (um)	f_p (%)
TiN-3	2	0.32	0.7
	1.4	5.3	0.2
TiN-5	2	0.31	1.6
	1.4	4.9	0.3

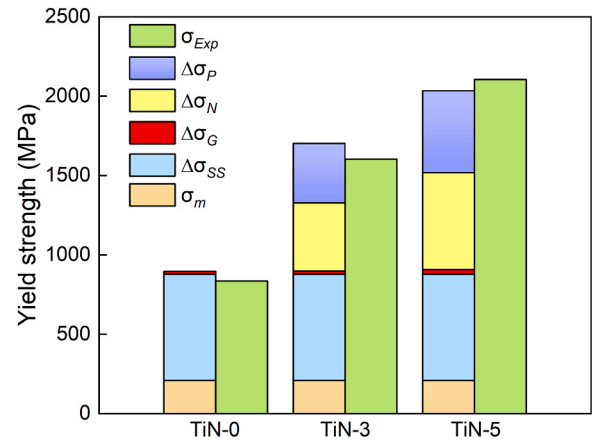


Fig. 8. Percentage contribution of different strengthening mechanism to the yield stress of TiZrNbVAl LHEA with different TiN contents.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.jallcom.2024.178063](https://doi.org/10.1016/j.jallcom.2024.178063).

Data availability

Data will be made available on request.

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